

Preventive Maintenance Grounding (PE, LE, FE, IE) (Examples/References)



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E&I Maintenance

PT. Multimas Nabati Asahan - Serang

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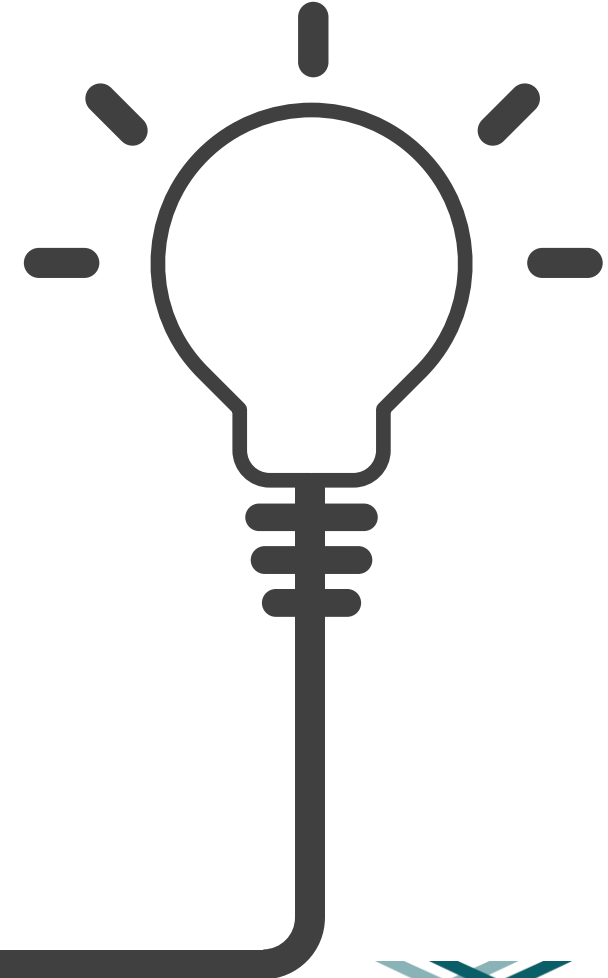
Preventive Maintenance (PM) Grounding

Grounding



Grounding adalah titik referensi dalam rangkaian listrik yang mana tegangan diukur, jalur balik yang sama untuk arus listrik, atau koneksi fisik langsung ke bumi *Grounding is a reference point in an electrical circuit from which a voltage is measured, a common return path for electric current, or a direct physical connection to earth.*

Tujuan utama pentanahan adalah untuk menghilangkan beda potensial dengan mengalirkan arus ke tanah atau bumi seperti kebocoran arus listrik atau sambaran petir *The main purpose of grounding is to eliminate potential differences by flowing the current to the ground or earth such as electric current leakage or lightning strike.*



Tujuan Sistem Grounding *The Objective of a Grounding System*

To assure that a person in the vicinity of grounded facilities is not exposed to the danger of critical electric shock

01

To provide means to dissipate electric currents into the earth without exceeding any operating and equipment limits

02

To prevent damage to electrical or electronic apparatus and dissipate lightning strokes

03

To assure correct operation of electrical or electronic device

04

To stabilize voltage during transient conditions and to minimize the probability of flashover during transients

06

To divert stray RF energy from sensitive audio, video, control, and computer equipment

07

Jenis-Jenis Sistem Grounding *Types of Grounding System*

PE

IE

FE

LE

**Protection Earth
Grounding**



**Instrumentation
Earth Grounding**



**Functional Earth
Grounding**

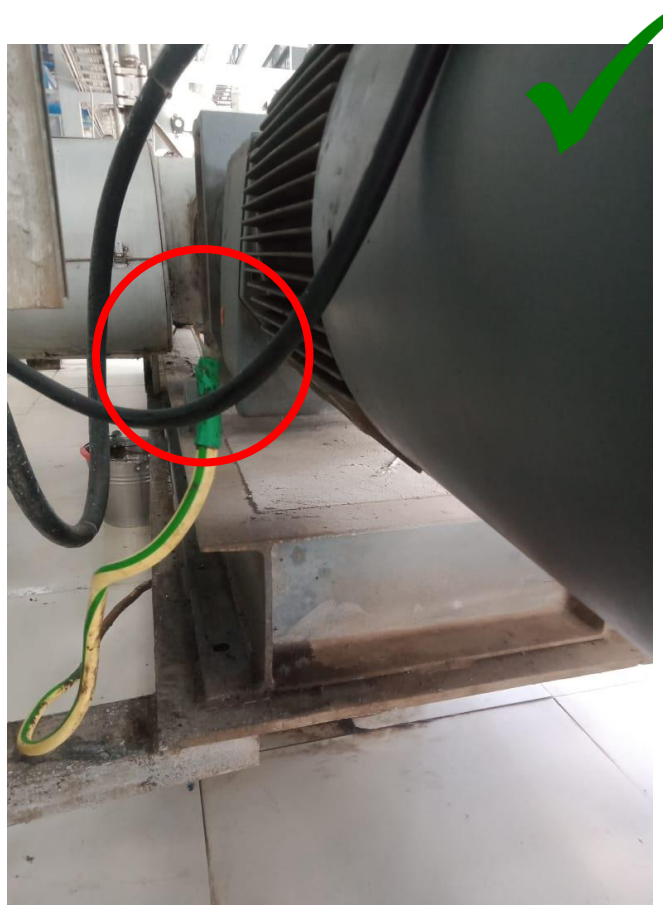


**Lightning Earth
Grounding**



Protection Earth (PE) Grounding

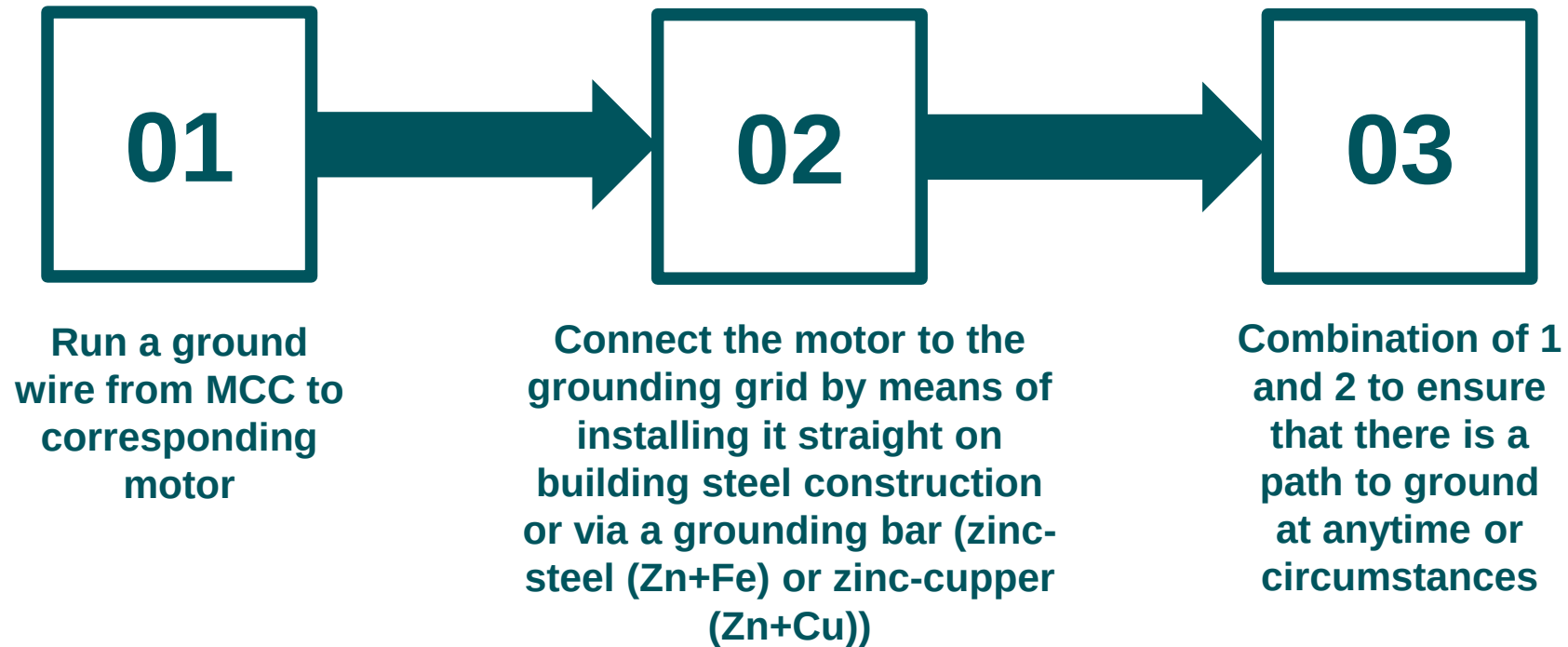
A. Motor Safety Grounding



The main reason for safety Protection Earth (PE) grounding is to create a path to ground. In case there is a fault, breaking or opening corresponding circuit breaker to prevent electrical shock hazard.

A. Motor Safety Grounding

There are three methods to accomplish a good motor grounding, hereby option 3 is preferred



B. Transformer Safety Grounding



Every piece of metal inside HV/MV transformer rooms/areas must be grounded properly

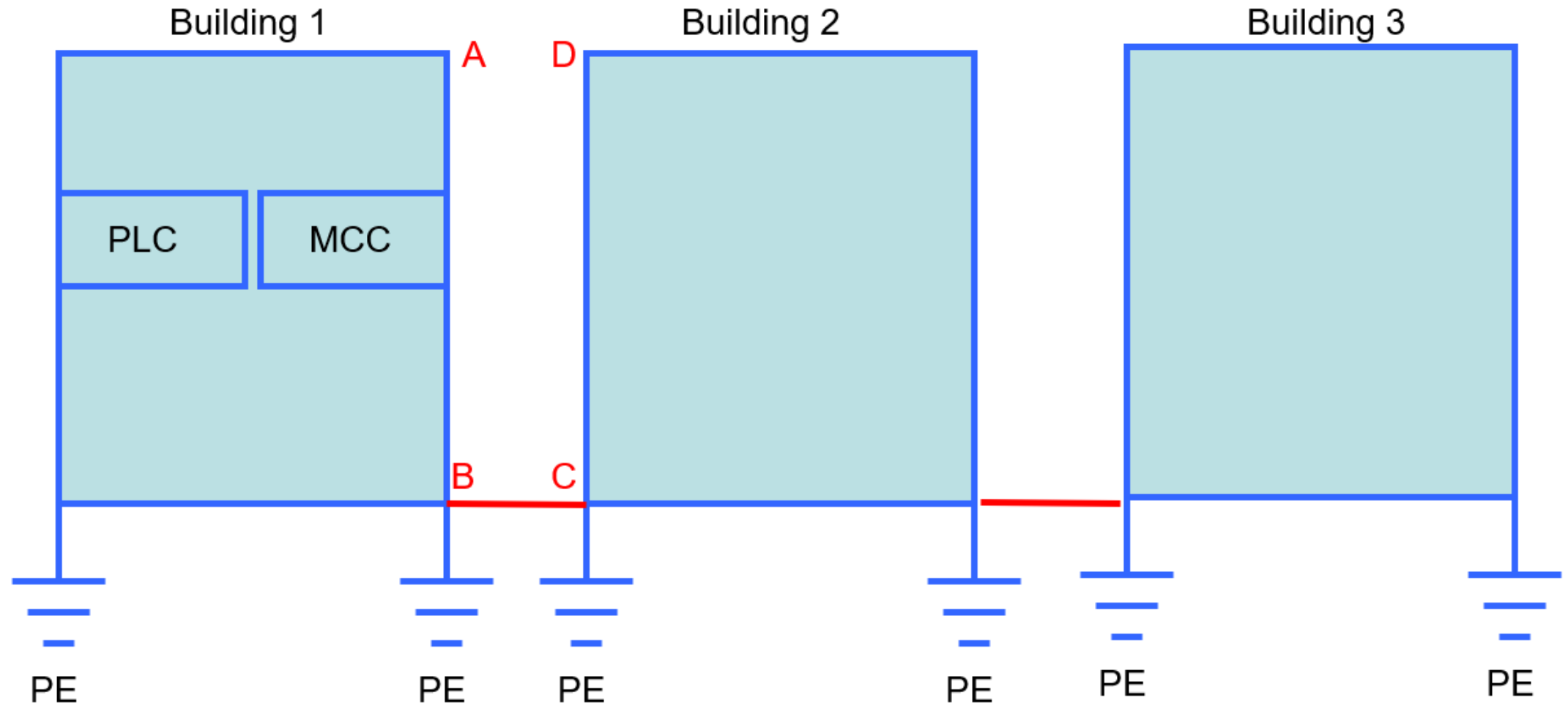


Good, solid ground strapping



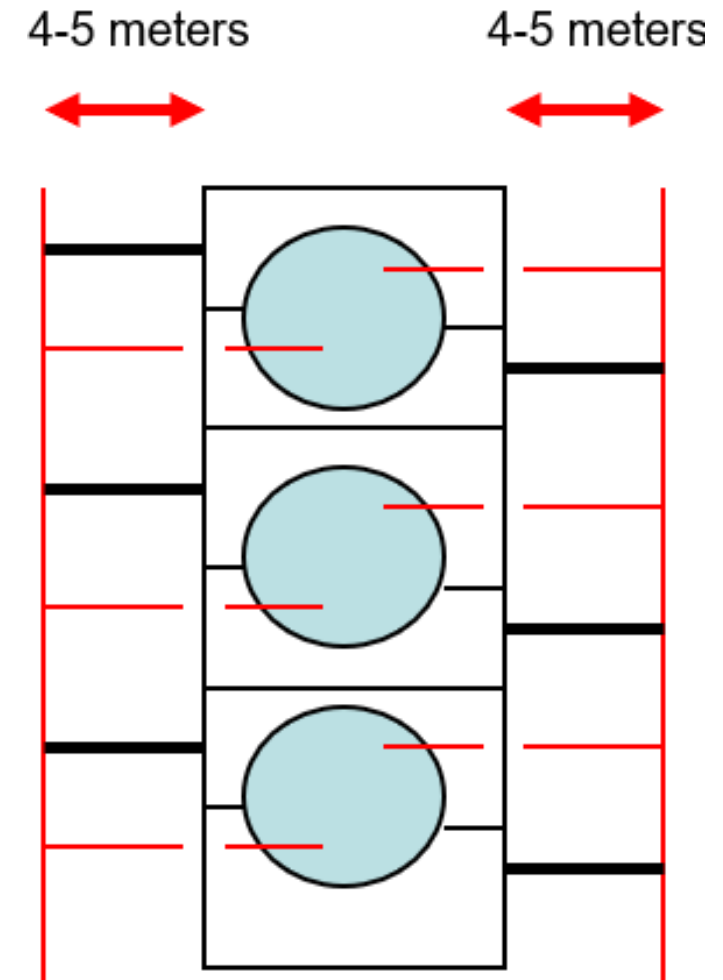
Broken or missing ground straps.

C. Building Safety Grounding

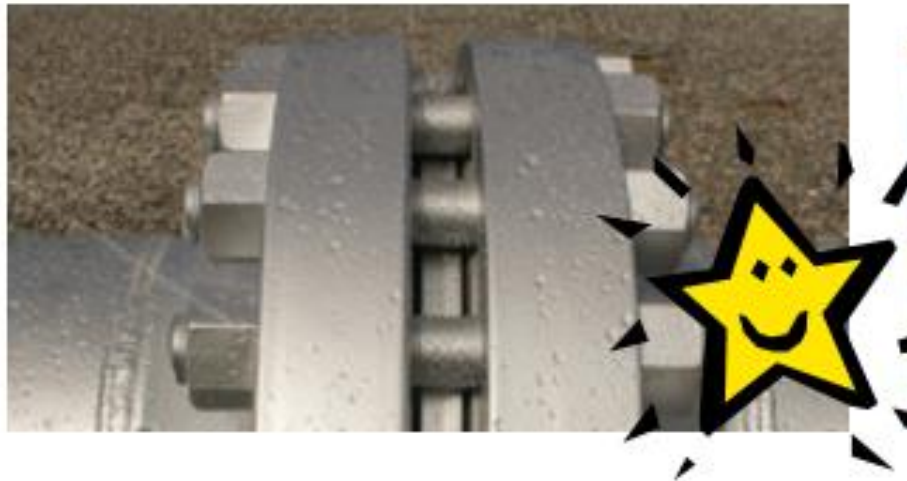


D. Tank-Farm Safety Grounding

Normally, we should create a grid around all our tanks, having each tank connected to this grid by at least 2 points (see illustration shown; black = protection/tank earth (PE); red = lightning earth (LE))

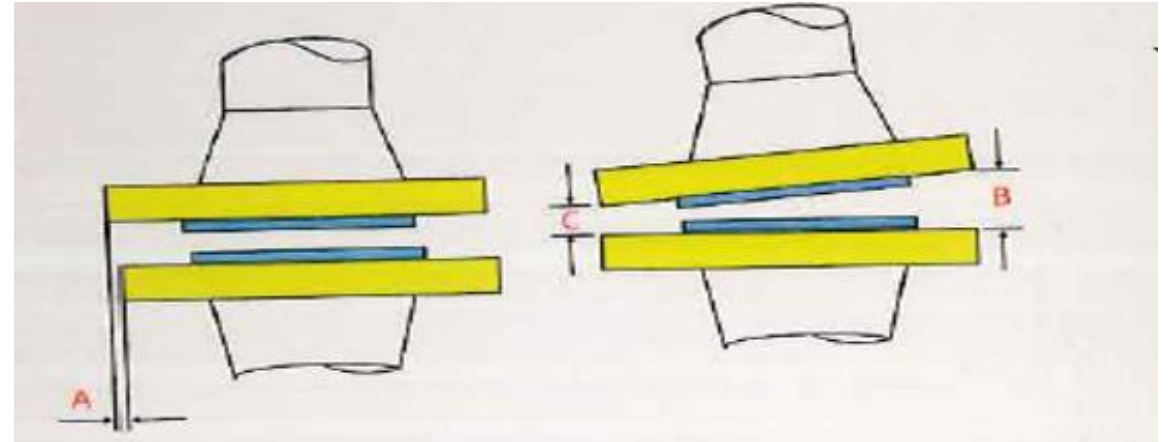


E. Pipping Safety Grounding



Pipe flanges to be properly bolted-down, aiming long-term trouble free operation.

All pipe, pipe's flange and applied bolts, nuts and washers need to be electrically bonded or connected to ground and properly tightened



Instrumentation Earth (IE) Grounding

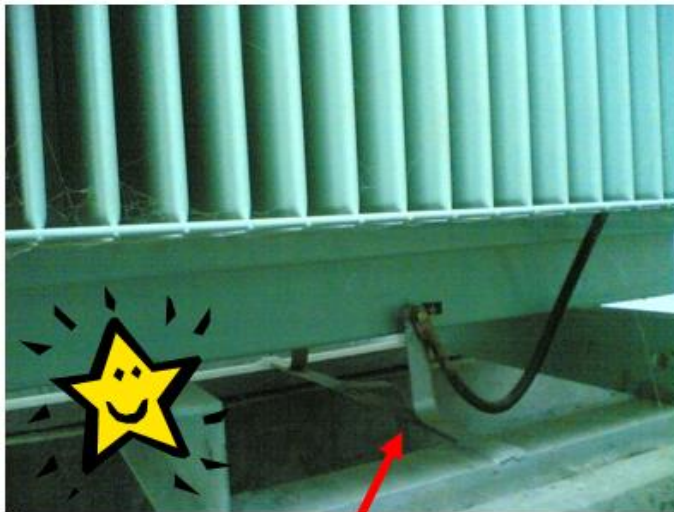
Instrumentation Earth (IE) Grounding

- A separate grounding rod should be used as Instrumentation Grounding (IG). This grounding rod should be installed 4 to 5 meters away from the building nearest Grounding Grid (GG) point.
- The IG should have two connection:
 1. To the overall grounding grid/system (it is very important in the event of major fault that everything floats together. By having this rod linked to the ground grid, we will accomplish this goal; at the same time noise in the ground grid will be significantly dissipated before it can have an effect on our instruments and computers)
 2. Straight to the DCS/PLC control system panels (creating a separate grounding path for instruments)

Functional Earth (FE) Grounding

Functional Earth (FE) Grounding

Functional Earth (FE) grounding is used to establish a good and stable point for neutral (N) and DC commons (0 VDC). Functional earth grounding is one of the most important connections in plants and therefore it's necessary to ensure that the connections are properly attached to corresponding grounding system



Good, solid ground strapping all the way to the grounding grid.

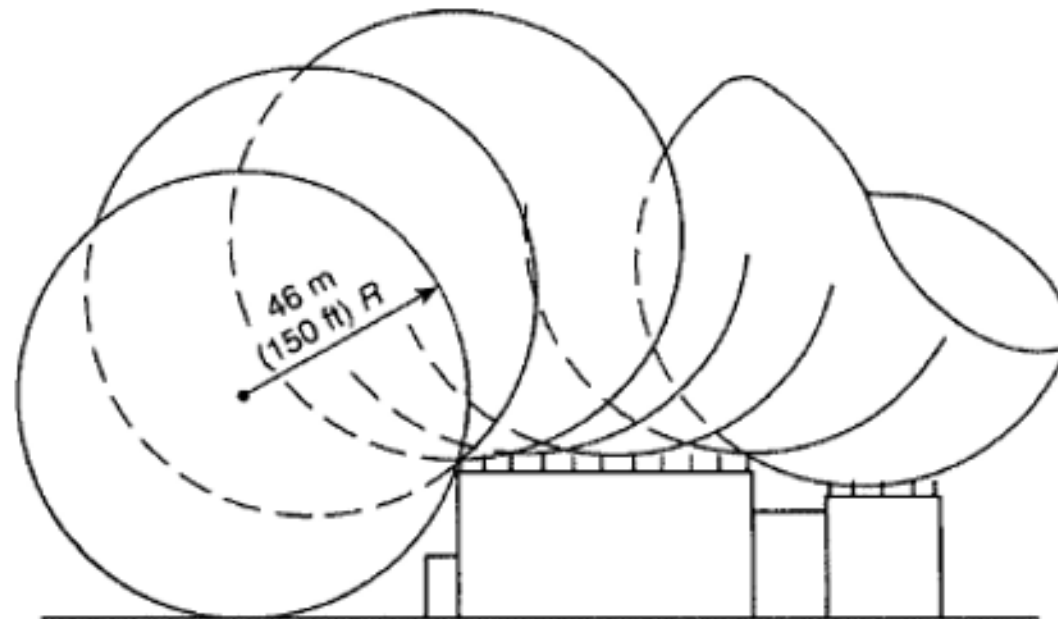


Good, solid ground strapping also, but not all the way to the grounding grid as we are using the metal construction beam as intermediate

Lightning Earth (LE) Grounding

Lightning Earth (LE) Grounding

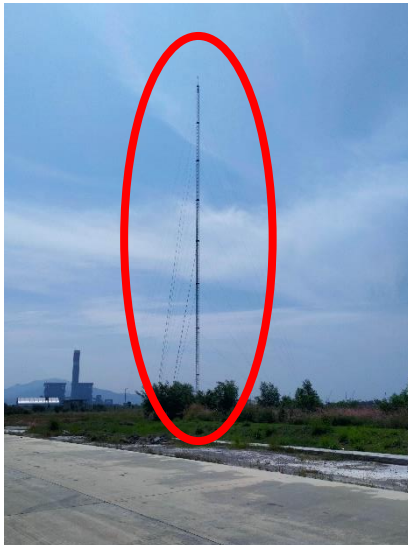
- A lightning hit into one of building or equipment could damage equipment and interfere its performance, due to high voltage surges during a very short period of time (μS)
- The lightning protection consists of 1) an antenna, to extract the lightning hit, and 2) a grounding cable, which is isolated from corresponding object that needs to be protected against lightning hits, pulled all the way to the ground or lowest point to ground of corresponding object



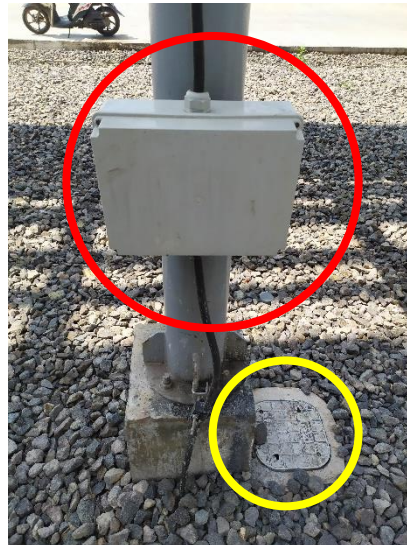
Lightning Earth (LE) Grounding



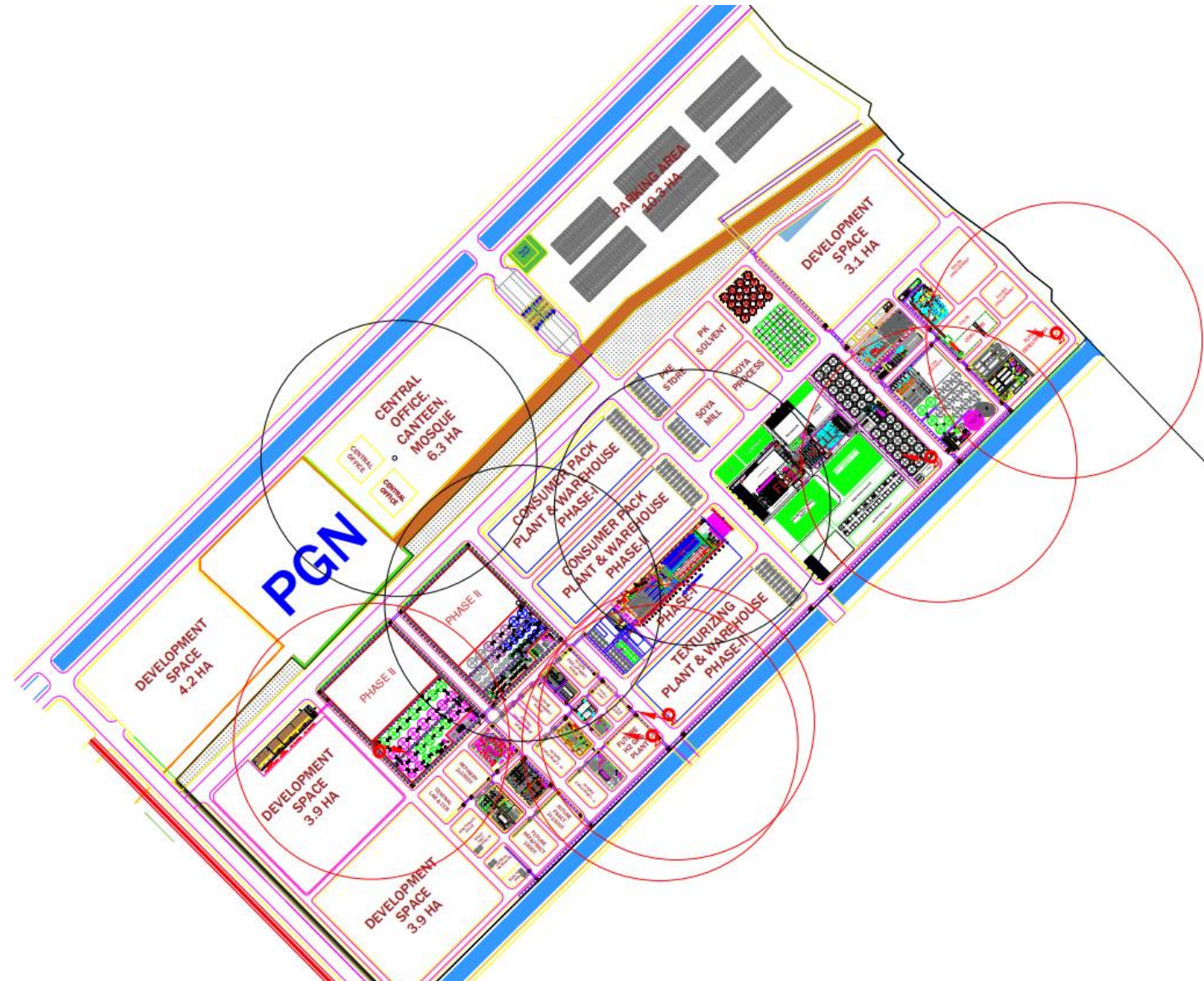
Building



Independent/
Stand Alone



Dispatcher LE





- = Zn+Fe or Zn+Cu bar to prevent it from rusting and/or corroding
- FE = Functional Earth
 - GG = (common) Ground Grid
 - IE = Instrumentation Earth
 - IG = Instrumentation Ground
 - LE = Lightning Earth
 - PE = Protection Earth

Lightning Earth (LE):

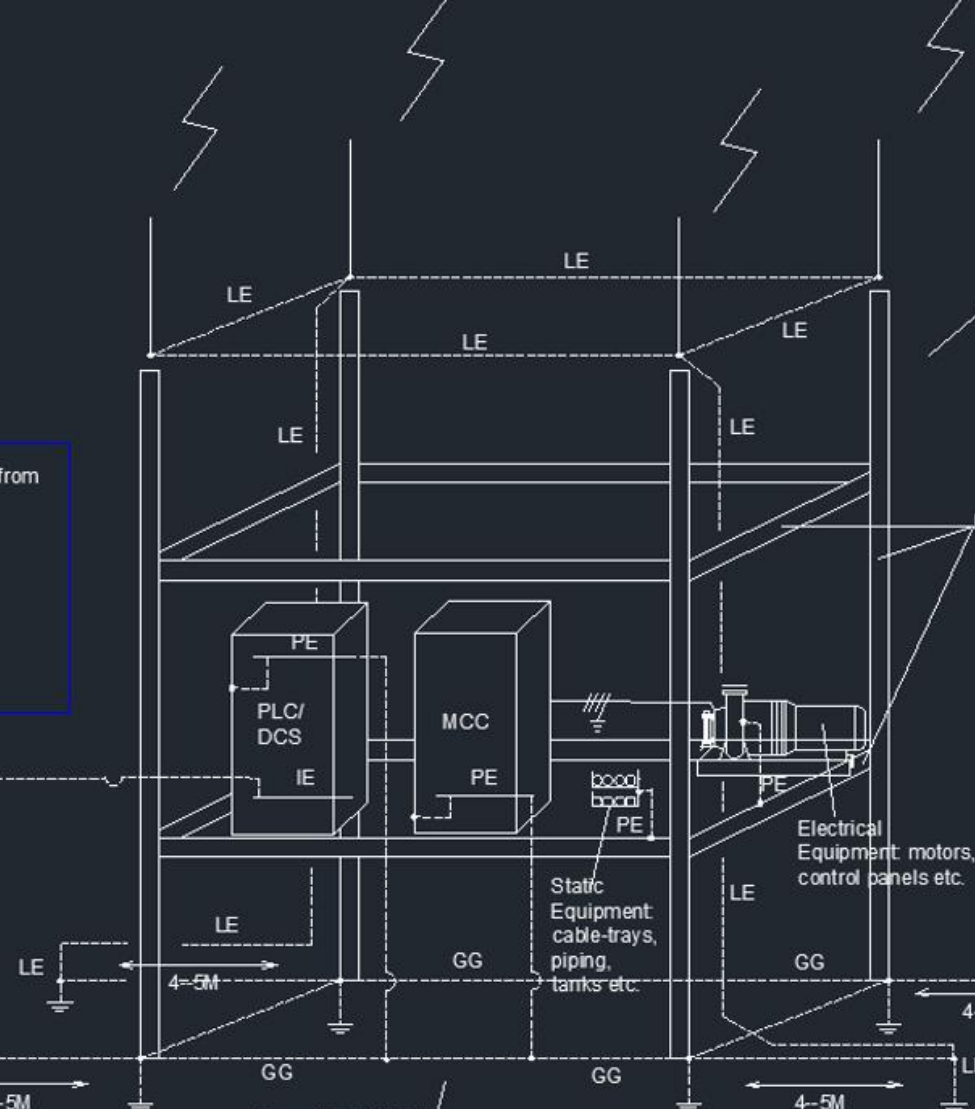
An separate grounding rod will be used as Lightning Earth (LE). The grounding rod must be installed 4 to 5 meters away from the nearest Ground Grid (GG) point.

Impedance must be smaller than or equal to 5-ohm, otherwise extra grounding rods will be installed.

Instrumentation Ground (IG):

An separate grounding rod will be used as Instrumentation Ground (IG). The grounding rod must be installed 4 to 5 meters away from the nearest Ground Grid (GG) point.

Impedance must be smaller than or equal to 1-ohm, otherwise extra grounding rods will be installed.



Ground Grid (GG):

Piling bar steel grounding by connecting the grounding zinc steel bar to the bar steel ring inside the foundation.

All ground connections to columns will be thermal (MIG (Metal Inert Gas)) welded.

Impedance must be smaller than or equal to 5-ohm, otherwise extra grounding rods will be installed.

Lightning Earth (LE):

An separate grounding rod will be used as Lightning Earth (LE). The grounding rod must be installed 4 to 5 meters away from the nearest Ground Grid (GG) point.

Impedance must be smaller than or equal to 5-ohm, otherwise extra grounding rods will be installed.

Lightning Earth (LE)

The lightning protection consists of 1) an antenna, to extract the lightning hit, and 2) a grounding cable, which is isolated from corresponding object that needs to be protected against lightning hits.

Building Steel



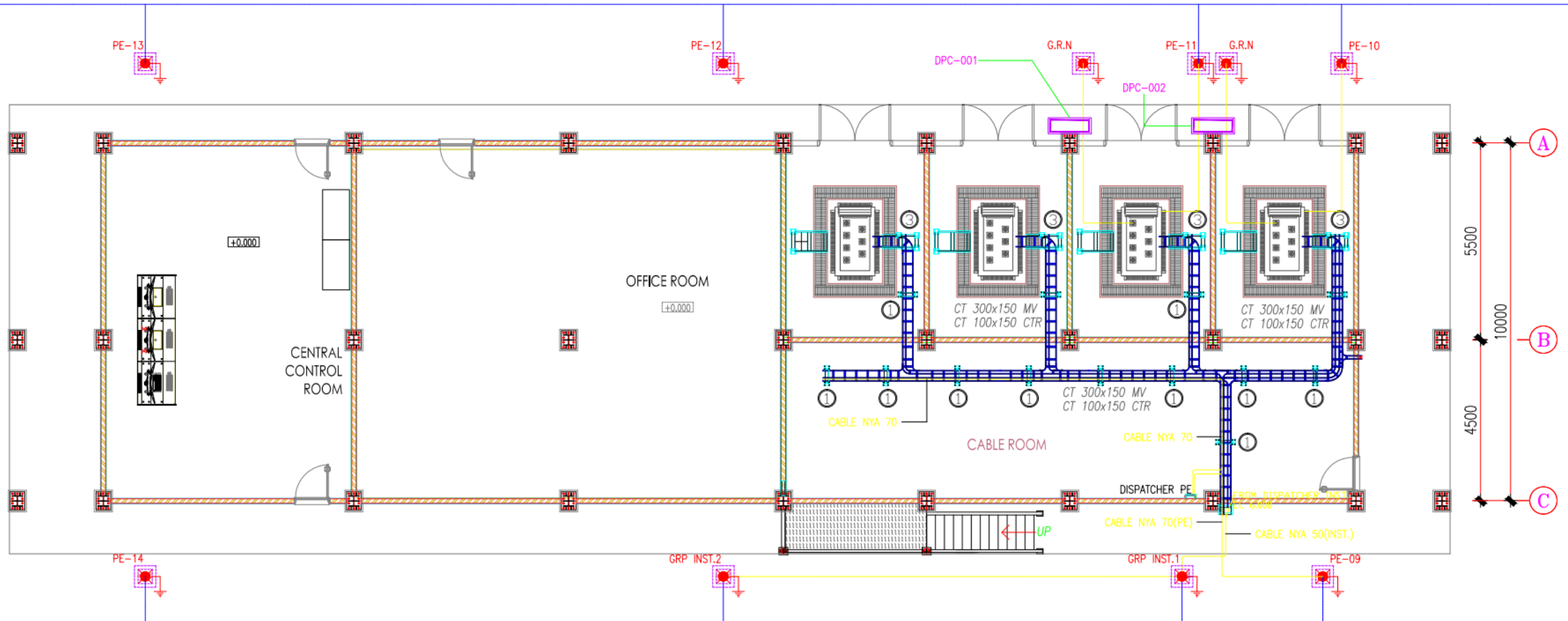
Functional Earth (FE):

An separate grounding rod will be used as Functional Earth (FE). The grounding rod must be installed 4 to 5 meters away from the nearest Ground Grid (GG) point.

Impedance must be smaller than or equal to 5-ohm, otherwise extra grounding rods will be installed.



Contoh Layout Sistem Grounding *Example of Grounding System Layout*



Standard Resistansi Sistem Grounding *Resistance Standard of Each Grounding System*

- The standard for measuring grounding resistance of each type based on the E&I Book can be seen in the table below

	PE	IE	FE	LE
Resistance (ohm)	<5	<1	<5	<5

Preventive Maintenance (PM) Grounding

FLUKE 1623-2 – Earth/Ground Tester GEO



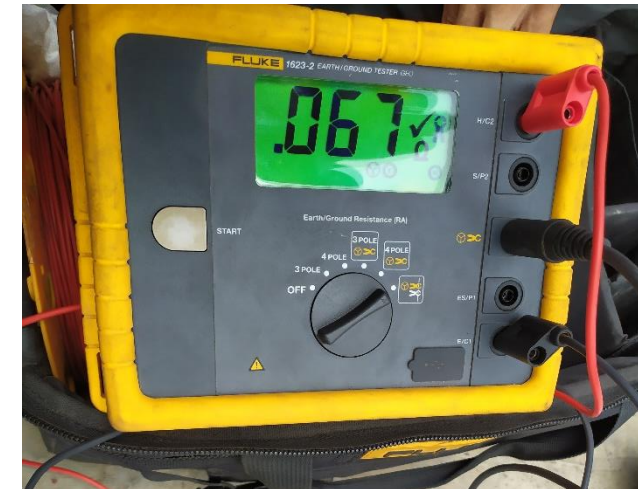
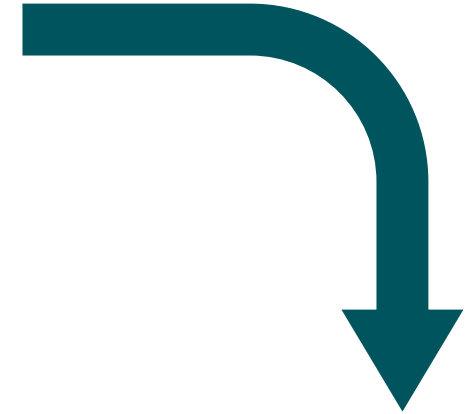
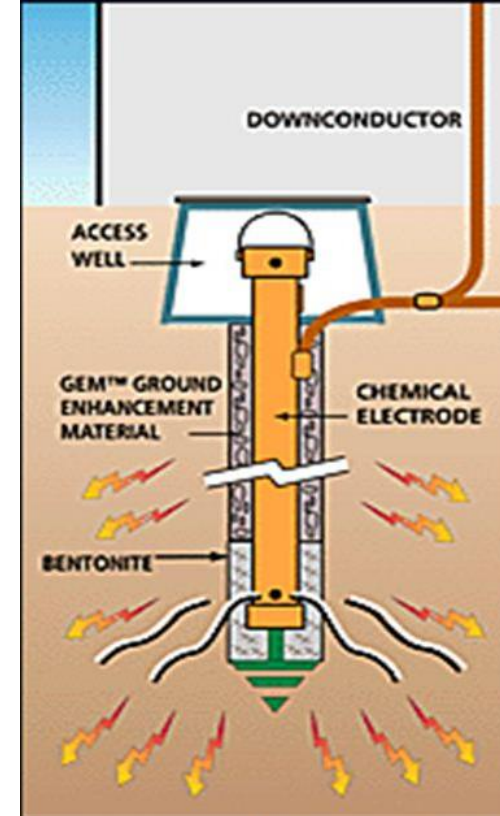
FLUKE 1630-2 – Earth Ground Clamp



	A	B	C	D	E	F	G	H	I
10	No	Tagging	Location	Picture 1	Picture 2	Picture 3		Remarks	
11	1	GRP - 001 - P	Fraksinasi	 0,536	 0,536	 0,502	<= 5 Ohm	Passed	
12	2	GRP - 002 - P	Fraksinasi	 0,483	 0,482	 0,483	<= 5 Ohm	Passed	
13				 0,560	 0,560	 0,560			

12	GRP - 001 - I (Grounding 3 PLC)	MCC	 0.111	 0.111	 0.111	$\leq 1 \text{ Ohm}$	Passed
13	GRP - 006 - P (Grounding 4 LVD 50R)	MCC	 9,7	 9,71	 9,69	$\geq 5 \text{ Ohm}$	#VALUE!
14	GRP - 007 - P (Grounding 5 Weater Heater)	MCC	 8.85	 8.50	 8.41	$\leq 5 \text{ Ohm}$	Passed

Corrective Action



Electrical Integrity Check



Electrical Integrity Check



